



深圳技术大学
SHENZHEN TECHNOLOGY UNIVERSITY

机器学习 -数学预备知识

温铁祥

健康与环境工程学院

wentiexiang@sztu.edu.cn

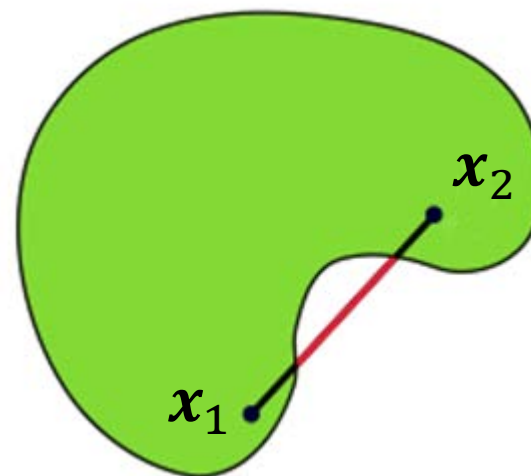
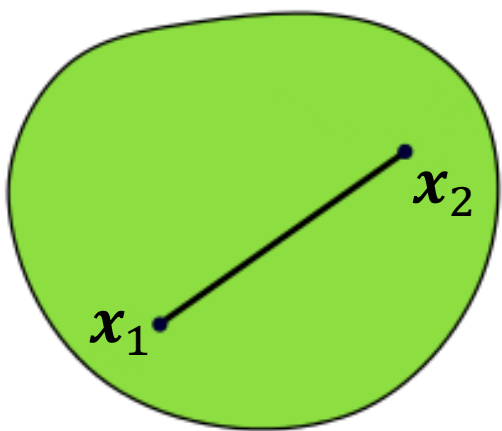


□ 凸集 (convex set)

对集合C内的任意两点 $x_1, x_2 \in C$, 若它们之间的连线上的所有点仍属于集合C, 即

$$\theta x_1 + (1 - \theta)x_2 \in C, \quad (\forall 0 \leq \theta \leq 1)$$

则我们称集合C为凸的, 即C是一个**凸集**。

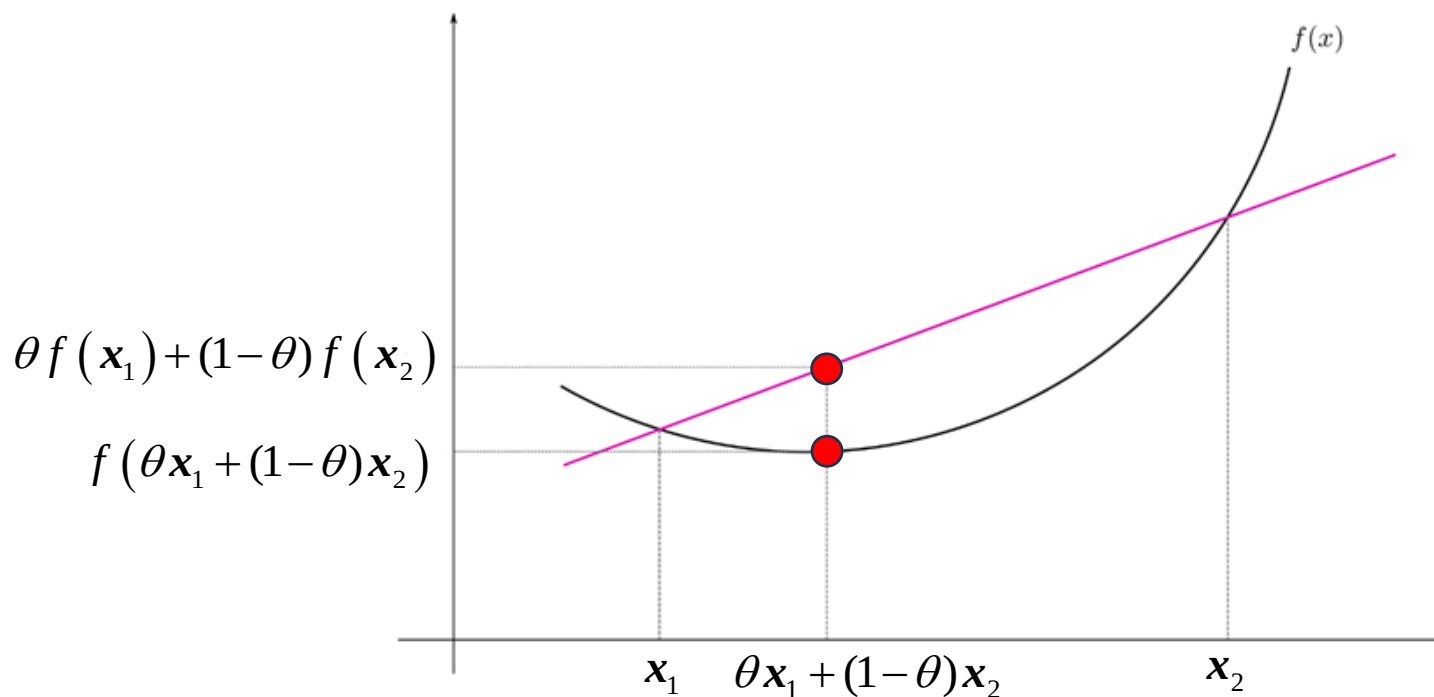


□ 凸函数 (convex function)

对定义在凸集 D 上的函数 $f: R^d \rightarrow R$, 若 $\forall x_1, x_2 \in D$ 均满足

$$f(\theta x_1 + (1-\theta)x_2) \leq \theta f(x_1) + (1-\theta)f(x_2) \quad (\forall 0 \leq \theta \leq 1)$$

则称函数 $f(\cdot)$ 为凸的, 即 $f(\cdot)$ 是一个**凸函数**。



□ 詹森不等式 (Jensen's inequality)

对定义的凸函数, 有

$$f\left(\sum_i \alpha_i x_i\right) \leq \sum_i \alpha_i f(x_i), \quad f(E_X[X]) \leq E_X[f(X)], \quad \text{其中} \sum_i \alpha_i = 1$$

凸函数的期望不小于期望的凸函数!

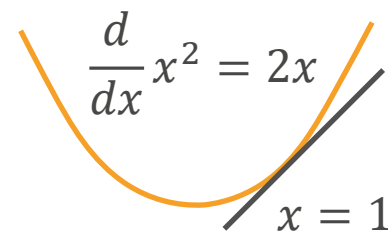
- 《动手学深度学习》第11.2小节

□ 标量求导回顾

y	a	x^n	$\exp(x)$	$\ln(x)$	$\sin(x)$
$\frac{dy}{dx}$	0	nx^{n-1}	$\exp(x)$	$\frac{1}{x}$	$\cos(x)$

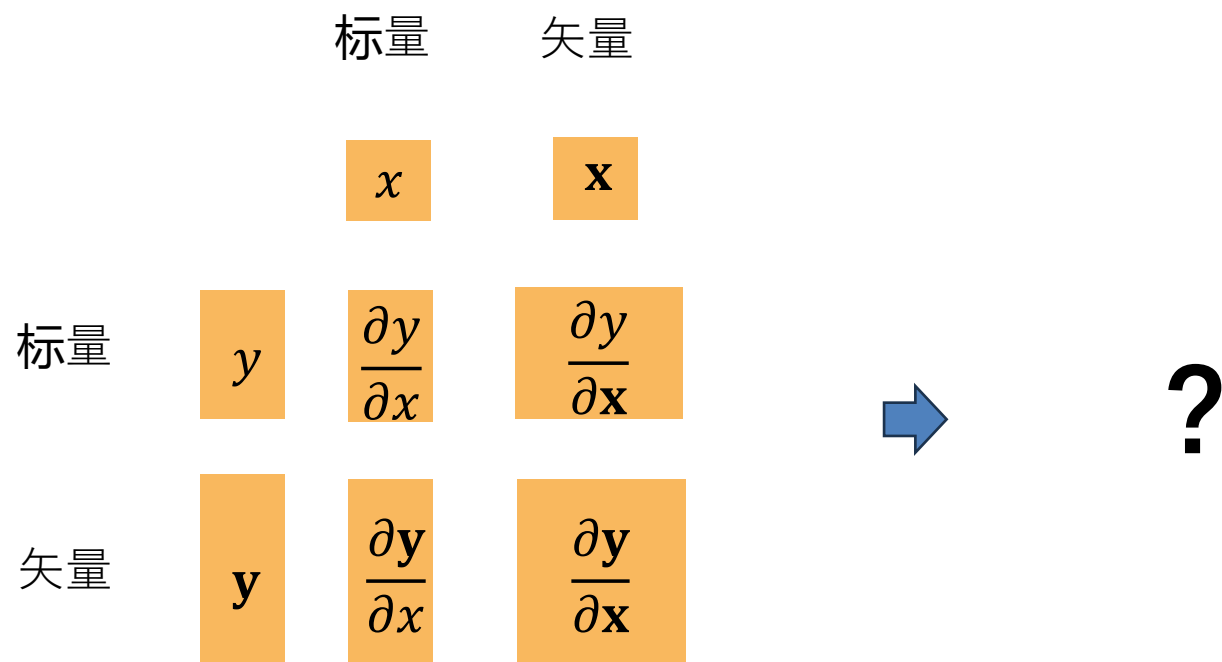
y	$u + v$	uv	$y = f(u), u = g(x)$
$\frac{dy}{dx}$	$\frac{du}{dx} + \frac{dv}{dx}$	$\frac{du}{dx}v + \frac{dv}{dx}u$	$\frac{dy}{du} \frac{du}{dx}$

导数是切线的斜率



切线的斜率为2

□ 梯度：矢量求导推广



□ 标量方程对向量的导数

$$y = f(x)$$

$$\frac{dy}{dx} = 0$$



$$y = f(x_1, x_2)$$
$$\begin{cases} \frac{\partial f(x_1, x_2)}{\partial x_1} = 0 \\ \frac{\partial f(x_1, x_2)}{\partial x_2} = 0 \end{cases}$$



$$y = f(x_1, x_2, \dots, x_n)$$

$$\begin{cases} \frac{\partial f(x_1, x_2, \dots, x_n)}{\partial x_1} = 0 \\ \frac{\partial f(x_1, x_2, \dots, x_n)}{\partial x_2} = 0 \\ \vdots \end{cases}$$

□ 标量方程对向量的导数

$$y = f(x_1, x_2, \dots, x_n)$$

$$\begin{cases} \frac{\partial f(x_1, x_2, \dots, x_n)}{\partial x_1} = 0 \\ \frac{\partial f(x_1, x_2, \dots, x_n)}{\partial x_2} = 0 \\ \vdots \end{cases}$$



$$y = f(\vec{x})$$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\frac{\partial f(\vec{x})}{\partial \vec{x}} = \begin{bmatrix} \frac{\partial f(\vec{x})}{\partial x_1} \\ \frac{\partial f(\vec{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\vec{x})}{\partial x_n} \end{bmatrix}_{n \times 1}$$

分母布局 (Denominator Layout 或 Hessian formulation)

□ 标量方程对向量的导数

$$y = f(x_1, x_2, \dots, x_n)$$

$$y = f(\vec{x})$$

$$\begin{cases} \frac{\partial f(x_1, x_2, \dots, x_n)}{\partial x_1} = 0 \\ \frac{\partial f(x_1, x_2, \dots, x_n)}{\partial x_2} = 0 \\ \vdots \end{cases}$$



$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\frac{\partial f(\vec{x})}{\partial \vec{x}} = \begin{bmatrix} \frac{\partial f(\vec{x})}{\partial x_1} & \frac{\partial f(\vec{x})}{\partial x_2} & \dots & \frac{\partial f(\vec{x})}{\partial x_n} \end{bmatrix}_{1 \times n}$$

分子布局 (Numerator Layout 或 Jacobian formulation)

$$\text{例: } f(x_1, x_2) = x_1^2 + x_2^2$$

□ 标量方程对向量的导数

- 常用特例1

$$\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1} \quad \vec{\beta} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_n \end{bmatrix}_{n \times 1}$$



$$\frac{\partial \vec{\beta}^T \vec{x}}{\partial \vec{x}} = \vec{\beta}$$

$$\frac{\partial \vec{x}^T \vec{\beta}}{\partial \vec{x}} = \vec{\beta}$$

分母布局

$$\frac{\partial \vec{\beta}^T \vec{x}}{\partial \vec{x}} = \vec{\beta}^T \quad \frac{\partial \vec{x}^T \vec{\beta}}{\partial \vec{x}} = \vec{\beta}^T$$

分子布局

□ 标量方程对向量的导数

例: $y = \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$

$$\vec{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$



$$\frac{\partial \vec{\beta}^T \vec{x}}{\partial \vec{x}} = \begin{bmatrix} \frac{\partial \beta^T \vec{x}}{\partial x_1} \\ \frac{\partial \beta^T \vec{x}}{\partial x_2} \\ \frac{\partial \beta^T \vec{x}}{\partial x_3} \end{bmatrix} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} = \vec{\beta}$$

$$\frac{\partial \vec{x}^T \vec{\beta}}{\partial \vec{x}} = \frac{\partial \vec{\beta}^T \vec{x}}{\partial \vec{x}} = \frac{\partial y}{\partial \vec{x}} = \vec{\beta}$$

分母布局

□ 向量方程对向量的导数

$$\begin{aligned}
 \vec{f}(\vec{x}) &= \begin{bmatrix} f_1(\vec{x}) \\ f_2(\vec{x}) \\ \vdots \\ f_m(\vec{x}) \end{bmatrix}_{m \times 1} \quad \rightarrow \quad \frac{\partial \vec{f}(\vec{x})_{m \times 1}}{\partial \vec{x}_{n \times 1}} = \begin{bmatrix} \frac{\partial \vec{f}(\vec{x})}{\partial x_1} \\ \frac{\partial \vec{f}(\vec{x})}{\partial x_2} \\ \vdots \\ \frac{\partial \vec{f}(\vec{x})}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1(\vec{x})}{\partial x_1} & \frac{\partial f_2(\vec{x})}{\partial x_1} & \dots & \frac{\partial f_m(\vec{x})}{\partial x_1} \\ \frac{\partial f_1(\vec{x})}{\partial x_2} & \frac{\partial f_2(\vec{x})}{\partial x_2} & \dots & \frac{\partial f_m(\vec{x})}{\partial x_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_1(\vec{x})}{\partial x_n} & \frac{\partial f_2(\vec{x})}{\partial x_n} & \dots & \frac{\partial f_m(\vec{x})}{\partial x_n} \end{bmatrix}_{n \times m} \\
 \vec{x} &= \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1}
 \end{aligned}$$

分母布局 (Denominator Layout)

□ 向量方程对向量的导数

$$\begin{aligned}
 \vec{f}(\vec{x}) &= \begin{bmatrix} f_1(\vec{x}) \\ f_2(\vec{x}) \\ \vdots \\ f_m(\vec{x}) \end{bmatrix}_{m \times 1} \quad \Rightarrow \quad \frac{\partial \vec{f}(\vec{x})_{m \times 1}}{\partial \vec{x}_{n \times 1}} = \begin{bmatrix} \frac{\partial f_1(\vec{x})}{\partial \vec{x}} \\ \frac{\partial f_2(\vec{x})}{\partial \vec{x}} \\ \vdots \\ \frac{\partial f_m(\vec{x})}{\partial \vec{x}} \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1(\vec{x})}{\partial x_1} & \frac{\partial f_1(\vec{x})}{\partial x_2} & \dots & \frac{\partial f_1(\vec{x})}{\partial x_n} \\ \frac{\partial f_2(\vec{x})}{\partial x_1} & \frac{\partial f_2(\vec{x})}{\partial x_2} & \dots & \frac{\partial f_2(\vec{x})}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m(\vec{x})}{\partial x_1} & \frac{\partial f_m(\vec{x})}{\partial x_2} & \dots & \frac{\partial f_m(\vec{x})}{\partial x_n} \end{bmatrix}_{m \times n} \\
 \vec{x} &= \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1}
 \end{aligned}$$

分子布局 (Denominator Layout)

□ 向量方程对向量的导数

例：

$$\vec{f}(\vec{x}) = \begin{bmatrix} f_1(\vec{x}) \\ f_2(\vec{x}) \end{bmatrix} = \begin{bmatrix} x_1^2 + x_2^2 + x_3 \\ 2x_1 + x_3^2 \end{bmatrix}$$
$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \rightarrow \quad \frac{\partial \vec{f}(\vec{x})}{\partial \vec{x}} = \begin{bmatrix} \frac{\partial \vec{f}(\vec{x})}{\partial x_1} \\ \frac{\partial \vec{f}(\vec{x})}{\partial x_2} \\ \frac{\partial \vec{f}(\vec{x})}{\partial x_3} \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1(\vec{x})}{\partial x_1} & \frac{\partial f_2(\vec{x})}{\partial x_1} \\ \frac{\partial f_1(\vec{x})}{\partial x_2} & \frac{\partial f_2(\vec{x})}{\partial x_2} \\ \frac{\partial f_1(\vec{x})}{\partial x_3} & \frac{\partial f_2(\vec{x})}{\partial x_3} \end{bmatrix} = \begin{bmatrix} 2x_1 & 2 \\ 2x_2 & 0 \\ 1 & 2x_3 \end{bmatrix}_{3 \times 2}$$

分母布局 (Denominator Layout)

□ 向量方程对向量的导数

• 常用特例2

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}_{m \times n} \quad \vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1}$$



$$\frac{\partial A \vec{x}}{\partial \vec{x}} = A^T$$

分母布局

$$\frac{\partial A \vec{x}}{\partial \vec{x}} = A$$

分子布局

□ 向量方程对向量的导数

例：

$$A\vec{x} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 \end{bmatrix}$$

$$\frac{\partial A\vec{x}}{\partial \vec{x}} = \begin{bmatrix} \frac{\partial A\vec{x}}{\partial x_1} \\ \frac{\partial A\vec{x}}{\partial x_2} \\ \frac{\partial A\vec{x}}{\partial x_3} \end{bmatrix} = \begin{bmatrix} \frac{\partial(a_{11}x_1 + a_{12}x_2 + a_{13}x_3)}{\partial x_1} & \frac{\partial(a_{21}x_1 + a_{22}x_2 + a_{23}x_3)}{\partial x_1} & \frac{\partial(a_{31}x_1 + a_{32}x_2 + a_{33}x_3)}{\partial x_1} \\ \frac{\partial(a_{11}x_1 + a_{12}x_2 + a_{13}x_3)}{\partial x_2} & \frac{\partial(a_{21}x_1 + a_{22}x_2 + a_{23}x_3)}{\partial x_2} & \frac{\partial(a_{31}x_1 + a_{32}x_2 + a_{33}x_3)}{\partial x_2} \\ \frac{\partial(a_{11}x_1 + a_{12}x_2 + a_{13}x_3)}{\partial x_3} & \frac{\partial(a_{21}x_1 + a_{22}x_2 + a_{23}x_3)}{\partial x_3} & \frac{\partial(a_{31}x_1 + a_{32}x_2 + a_{33}x_3)}{\partial x_3} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix} = A^T$$

□ 向量方程对向量的导数

• 常用特例3

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}_{m \times n} \quad \vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1}$$



$$\frac{\partial \vec{x}^T A \vec{x}}{\partial \vec{x}} = A \vec{x} + A^T \vec{x}$$

$$\frac{\partial \vec{x}^T A \vec{x}}{\partial \vec{x}} = 2A \vec{x}, \text{ 若 } A \text{ 对称}$$

分母布局

□ 向量方程对向量的导数

例：

$$\vec{x}^T A \vec{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{bmatrix} = x_1(a_{11}x_1 + a_{12}x_2) + x_2(a_{21}x_1 + a_{22}x_2)$$

$$\frac{\partial \vec{x}^T A \vec{x}}{\partial \vec{x}} = \begin{bmatrix} \frac{\partial \vec{x}^T A \vec{x}}{\partial x_1} \\ \frac{\partial \vec{x}^T A \vec{x}}{\partial x_2} \end{bmatrix}_{2 \times 1} = \begin{bmatrix} (a_{11}x_1 + a_{12}x_2) + a_{11}x_1 + a_{21}x_2 \\ a_{12}x_1 + (a_{21}x_1 + a_{22}x_2) + a_{22}x_2 \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{bmatrix} + \begin{bmatrix} a_{11}x_1 + a_{21}x_2 \\ a_{12}x_1 + a_{22}x_2 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = A \vec{x} + A^T \vec{x}$$

□ 向量方程对向量的导数

例：已知二次型

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \vec{x}^T A \vec{x} = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1^2 - 3x_3^2 - 4x_1x_2 + x_2x_3$$

$$\frac{\partial \vec{x}^T A \vec{x}}{\partial \vec{x}} = \begin{bmatrix} \frac{\partial \vec{x}^T A \vec{x}}{\partial x_1} \\ \frac{\partial \vec{x}^T A \vec{x}}{\partial x_2} \\ \frac{\partial \vec{x}^T A \vec{x}}{\partial x_3} \end{bmatrix} = \begin{bmatrix} 2x_1 - 4x_2 \\ -4x_1 + x_3 \\ -6x_3 + x_2 \end{bmatrix}$$

直接求解

$$\frac{\partial \vec{x}^T A \vec{x}}{\partial \vec{x}} = \begin{bmatrix} \frac{\partial \vec{x}^T A \vec{x}}{\partial x_1} \\ \frac{\partial \vec{x}^T A \vec{x}}{\partial x_2} \\ \frac{\partial \vec{x}^T A \vec{x}}{\partial x_3} \end{bmatrix} = 2A\vec{x} = 2 \begin{bmatrix} 1 & -2 & 0 \\ -2 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_1 - 4x_2 \\ -4x_1 + x_3 \\ x_2 - 6x_3 \end{bmatrix}$$

代公式

□ 向量方程对向量的导数

• 常用特例3

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}_{m \times n} \quad \vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1}$$



$$\frac{\partial \vec{x}^T A \vec{x}}{\partial \vec{x}} = A \vec{x} + A^T \vec{x}$$

$$\frac{\partial \vec{x}^T A \vec{x}}{\partial \vec{x}} = 2A \vec{x}, \text{ 若 } A \text{ 对称}$$

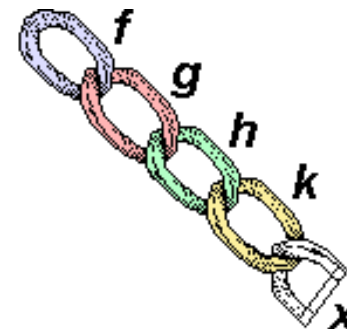
$$\frac{\partial \vec{x}^T \vec{x}}{\partial \vec{x}} = \frac{\partial \|\vec{x}\|_2^2}{\partial \vec{x}} = 2\vec{x}, \text{ 若 } A=I \text{ 单位矩阵}$$

分母布局

□ 链式法则 (chain rule)

- 链式法则 - 标量:

$$y = f(u), u = g(x) \quad \frac{\partial y}{\partial x} = \frac{\partial y}{\partial u} \frac{\partial u}{\partial x}$$



- 链式法则 - 矢量:

分母布局:

(“前乘”)

$$\frac{\partial y}{\partial \mathbf{x}} = \frac{\partial u}{\partial \mathbf{x}} \frac{\partial y}{\partial u}$$

$$(n, 1) \quad (n, 1) (1, 1)$$

$$\frac{\partial y}{\partial \mathbf{x}} = \frac{\partial \mathbf{u}}{\partial \mathbf{x}} \frac{\partial y}{\partial \mathbf{u}}$$

$$(n, 1) \quad (n, k) (k, 1)$$

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \frac{\partial \mathbf{u}}{\partial \mathbf{x}} \frac{\partial \mathbf{y}}{\partial \mathbf{u}}$$

$$(n, m) \quad (n, k) (k, m)$$

分子布局:

(“后乘”)

$$\frac{\partial y}{\partial \mathbf{x}} = \frac{\partial y}{\partial u} \frac{\partial u}{\partial \mathbf{x}}$$

$$(1, n) \quad (1, 1) (1, n)$$

$$\frac{\partial y}{\partial \mathbf{x}} = \frac{\partial y}{\partial \mathbf{u}} \frac{\partial \mathbf{u}}{\partial \mathbf{x}}$$

$$(1, n) \quad (1, k) (k, n)$$

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \frac{\partial \mathbf{y}}{\partial \mathbf{u}} \frac{\partial \mathbf{u}}{\partial \mathbf{x}}$$

$$(m, n) \quad (m, k) (k, n)$$

□ 严格数学证明

- Randal J. Barnes, Matrix Differentiation.